Shiva Yeshlur

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EDUCATION

University of Minnesota – Twin Cities **Twin Cities, MN**
Ph.D., Mathematics *Expected May 2032*

Columbia University **New York City, NY**
B.A., Mathematics *May 2026*

- GPA: 3.92
- Coursework: Applied Machine Learning, Group Theory, Ring and Field Theory, Real Analysis, Measure Theory, Algebraic Topology, Differential Geometry, Graduate Algebraic Topology I, Graduate Algebraic Topology II

RESEARCH EXPERIENCE

AlQuraishi Lab, Columbia University Irving Medical Center **New York City, NY**
Research Assistant – Machine Learning *January 2024 - June 2026*

- Developed scalable graph-based clustering algorithms for benchmark dataset construction, improving state-of-the-art model accuracy by 12% on out-of-distribution protein-protein interaction tasks.
- Maintained distributed training pipelines on multi-node GPU clusters and implemented relevant DeepSpeed optimizations, enhancing pipeline throughput by 5%.
- Benchmarked transformer and GNN architectures; introduced a novel validation metric for measuring dataset–model distributional shifts given similarity thresholding of training data.
- Contributed benchmarking results to OpenFold, an open-source AlphaFold 2/3 implementation used by 10+ academic labs for protein complex modeling.

Columbia University Department of Mathematics **New York City, NY**
Senior Project, Advised by Lucy Yang *October 2025 - May 2026*

- Developed an expository paper on the Quillence equivalence between the homotopy category of rational, finite type, simply-connected spaces and the homotopy category of similar-adjective CDGAs, emphasizing the topological intuition of the original proof.
- Presented the exposition to 10+ graduate students in a graduate-run seminar as an undergraduate.

Columbia University Department of Mathematics **New York City, NY**
Summer Research Fellow – Mathematics *May 2024 - August 2024*

- Led a cross-functional project optimizing urban infrastructure networks in New York City.
- Leveraged OpenStreetMaps API data to generate complex graphical representations of Manhattan's traffic network with detailed point-of-interest information.
- Utilized effective graph resistance and Monte Carlo Markov Chain (MCMC) simulations to compare traffic congestion in the status quo and our modified network.
- Presented project outcomes at the Jane Street Math Day and Columbia Undergraduate Symposium, emphasizing real-world policy implementation for key stakeholders, including urban planners.

SKILLS

Programming: Python | Git | Bash
Machine Learning: Protein language models | Encoder-only architectures (e.g. ESM-2) | Representation Learning
High-Performance Computing: Multi-GPU training | DeepSpeed | Parallel computing | Slurm | PBS | Federal HPC systems (NERSC Perlmutter, ALCF Polaris) | Cloud computing

FELLOWSHIPS AND AWARDS

Kevin D. Eng & UnHae Song Foundation Fellow (2025) ($4860)
Columbia Summer Undergraduate Research Experience in Mathematical Modeling (2024) ($3500)
Rachelefsky Research Fellow (2024) ($2990)
Dean's List (2022 - Present)
National Rural and Small Town Scholar (2022)

PRESENTATIONS

Columbia Undergraduate Research Symposium, New York, NY, October 2024. **S. Yeshlur**; A. Chen; C. Pineda. "Reimagining Manhattan: A Mathematical Modeling Approach to Superblock Development." (poster)
Columbia Pathways Program Symposium, New York, NY, August 2024. **S. Yeshlur**; A. Chen; C. Pineda. "Reimagining Manhattan: A Mathematical Modeling Approach to Superblock Development." (poster)
Jane Street Math Day, New York, NY, July 2024. **S. Yeshlur**; A. Chen; C. Pineda. "Reimagining Manhattan: A Mathematical Modeling Approach to Superblock Development." (poster)

ACTIVITIES AND SERVICE

Mathematics Project of Minnesota, Mentor (Oct 2026 – Present) Organizing a 4-day workshop for mathematics undergraduates.
Rural Recruitment Committee, Member (Aug 2023 – May 2026) – Supported prospective rural student applicants, and helped organize the multi-day STARS Fly-In Program for promising students.
Columbia Political Review, Staff Writer, Junior Editor (Jan 2023 – Aug 2024) – Published an opinion article on the 2022 Sudanese coup, and edited 10+ staff writer pieces.